1. **Write and explain a C program to print the message "*Hello World*".**

Ans:-

Code:-

```
#include <stdio.h>

int main()

{

printf("Hello World!\n");

return 0;

}
```

<u>Output:-</u>

2. Write a C program to input an integer from the user and check whether the number is even or odd. Print the appropriate message.

Ans:-

```
#include <stdio.h>

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0) {
        printf("%d is Even.\n", num);
    } else {
        printf("%d is Odd.\n", num);
    }

    return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc even_or_oddt.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: 6
6 is Even.
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

3. Write a C program to check whether a given number is a prime number or not.

- **A prime number is a number greater than 1 that has no divisors other than 1 and itself.**
- **Take the number as input from the user and print whether it is prime or not prime.**

Ans:-

```
#include <stdio.h>

int main() {

int num, i, flag = 0;

printf("Enter a number: ");

scanf("%d", &num);

if (num <= 1)
```

```
{
        flag = 1;
}
for (i = 2; i <= num / 2; i++)
{
   if (num % i == 0)
        {
   flag = 1;
   break;
   }
   }

   if (flag == 0)
        {
        printf("%d is a Prime number.\n", num);
   }
        else
        {
   printf("%d is not a Prime number.\n", num);
   }
```

```
    return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: 6
6 is not a Prime number.
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: 5
5 is a Prime number.
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

4. Write a C program to check whether a given number is a palindrome or not.

Ans:-

```
#include <stdio.h>
int main()
{
int n,org, rev = 0, rem;
printf("Enter a number: ");
scanf("%d", &n);
```

```
org = n;
while (n != 0)
{
   rem = n % 10;
   rev = rev * 10 + rem;
   n /= 10;
}
if (org == rev)
 {
   printf("%d is a Palindrome number.\n", org);
}
else
{
   printf("%d is not a Palindrome number.\n", org);
}
        return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc palindrome_or_not.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: 156
156 is not a Palindrome number.
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: 55
55 is a Palindrome number.
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

5. Write a C program to check whether a given number is positive or negative

Ans:-

```
#include <stdio.h>


int main()

{

int n;

printf("Enter a number: ");

scanf("%d", &n);

if (n > 0)

   printf("Positive");

else if (n < 0)
```

```
    printf("Negative");
else
    printf("Zero");


return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc positive_negative.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: -6
Negativesrinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a number: 7
Positivesrinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

6. Write a C program to input a year from the user and check whether it is a leap year or not.

Ans:-

```
#include <stdio.h>
int main()
{
int year;
```

```
printf("Enter a year: ");
scanf("%d", &year);
if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
   printf("%d is a Leap Year", year);
else
   printf("%d is NOT a Leap Year", year);
return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-Not
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc leap_year.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a year: 2004
2004 is a Leap Yearsrinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter a year: 2026
2026 is NOT a Leap Yearsrinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

7. Write a C program to generate the Fibonacci series up to n terms.

Ans:-

```
#include <stdio.h>
int main()
{
int n, a = 0, b = 1, c, i;
printf("Enter number of terms: ");
scanf("%d", &n);
printf("Fibonacci Series: ");
for (i = 1; i <= n; i++)
{
printf("%d ", a);
c = a + b;
a = b;
b = c;
}
return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc fibonacci_series.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter number of terms: 7
Fibonacci Series: 0 1 1 2 3 5 8 srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

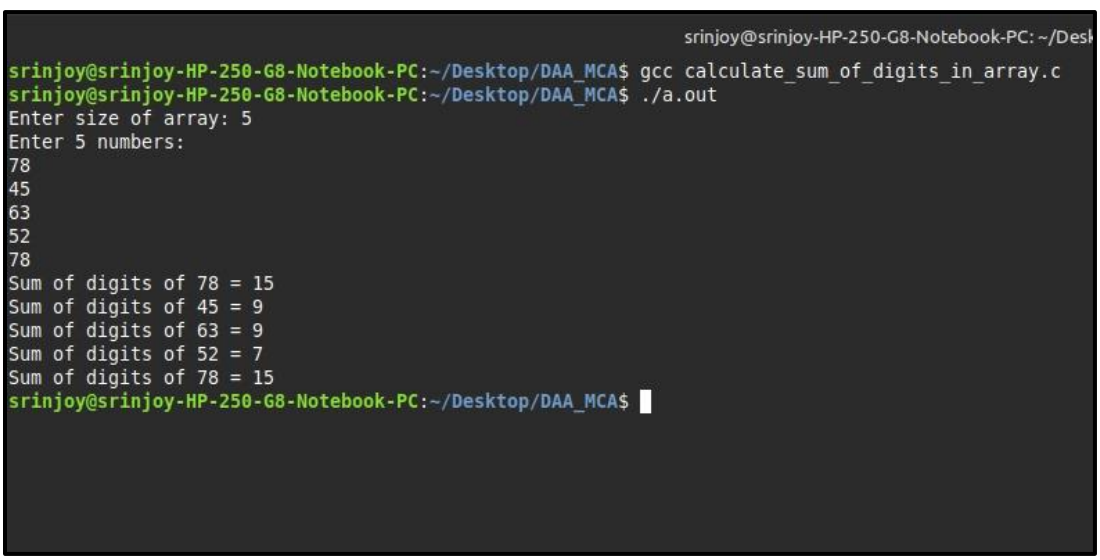
8. Write a C program to calculate the sum of digits of each element in an array.

Ans:-

```
#include <stdio.h>

int main()

{

int n, i, num, sum;

printf("Enter size of array: ");

scanf("%d", &n);

int a[n];

printf("Enter %d numbers:\n", n);

for (i = 0; i < n; i++)

    scanf("%d", &a[i]);

for (i = 0; i < n; i++)

{
```

```
    num = a[i];
    sum = 0;
        while (num)
        {
    sum += num % 10;
    num /= 10;
    }
        printf("Sum of digits of %d = %d\n", a[i], sum);
    }
return 0;
}
```

Output:-



9. Write a C program to find the maximum and minimum elements in an array.

Ans:-

```
#include <stdio.h>

int main()

{

int n, i;

printf("Enter size of array: ");

scanf("%d", &n);

int a[n];

printf("Enter %d numbers:\n", n);

for (i = 0; i < n; i++)

   scanf("%d", &a[i]);

int max = a[0], min = a[0];

for (i = 1; i < n; i++)

{

   if (a[i] > max) max = a[i];

   if (a[i] < min) min = a[i];

}

printf("Maximum = %d\nMinimum = %d", max, min);

return 0;

}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-No
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc maximum_and_minimum.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter size of array: 5
Enter 5 numbers:
45
52
56
89
78
Maximum = 89
Minimum = 45srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

10. Write a C program to search for a number in an array using Linear Search.

Ans:-

```
#include <stdio.h>

int main()

{

int n, i, key;

printf("Enter size of array: ");

scanf("%d", &n);

int a[n];

printf("Enter %d numbers:\n", n);

for (i = 0; i < n; i++)

   scanf("%d", &a[i]);
```

```
printf("Enter number to search: ");
scanf("%d", &key);
for (i = 0; i < n; i++)
 {
   if (a[i] == key)
        {
   printf("Number found at position %d\n", i + 1);
   return 0;
   }
   }
   printf("Number not found\n");
   return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc linear_search.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter size of array: 5
Enter 5 numbers:
4
2
5
6
9
Enter number to search: 6
Number found at position 4
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

11. Write a C program to search for a number in a sorted array using Binary Search.

Ans:-

```
#include <stdio.h>


int main()
{
   int n, i, key, low, high, mid;


   printf("Enter size of array: ");
   scanf("%d", &n);


   int a[n];  // Array declaration


   printf("Enter %d sorted numbers:\n", n);
   for (i = 0; i < n; i++)
      scanf("%d", &a[i]);


   printf("Enter number to search: ");
   scanf("%d", &key);


   low = 0;
```

```
    high = n - 1;

    // Binary Search
    while (low <= high)
    {
        mid = (low + high) / 2;

        if (a[mid] == key)
        {
            printf("Number found at position %d\n", mid + 1);
            return 0;
        }
        else if (a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }

    printf("Number not found\n");

    return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc binary_search.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Enter size of array: 8
Enter 8 sorted numbers:
78
56
74
52
63
69
80
90
Enter number to search: 63
Number found at position 5
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

12. Write a C program to sort an array of integers in ascending order using Merge Sort.

Ans:-

```
#include <stdio.h>

void merge(int arr[], int left, int mid, int right) {

    int n1 = mid - left + 1;

    int n2 = right - mid;

    int leftArr[n1], rightArr[n2];

    int i, j, k;

    for (i = 0; i < n1; i++)
```

```
    leftArr[i] = arr[left + i];
for (j = 0; j < n2; j++)
    rightArr[j] = arr[mid + 1 + j];



i = 0;
j = 0;
k = left;

while (i < n1 && j < n2) {
    if (leftArr[i] <= rightArr[j]) {
        arr[k] = leftArr[i];
        i++;
    } else {
        arr[k] = rightArr[j];
        j++;
    }
    k++;
}



while (i < n1) {
```

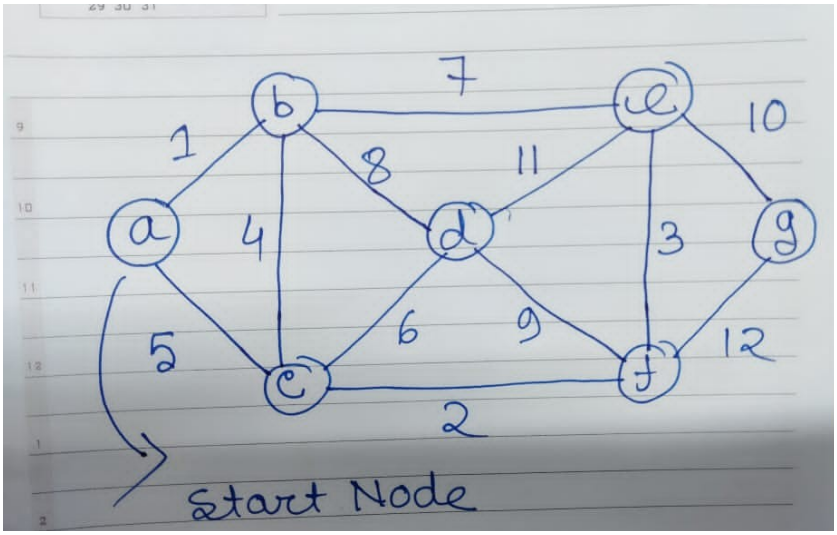
```
        arr[k] = leftArr[i];
        i++;
        k++;
    }
    while (j < n2) {
        arr[k] = rightArr[j];
        j++;
        k++;
    }
}


void mergeSort(int arr[], int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;



        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);



        merge(arr, left, mid, right);
```

```
    }
}

int main() {
    int arr[] = {38, 27, 43, 3, 9, 82, 10};
    int size = sizeof(arr) / sizeof(arr[0]);
    int i;

    printf("Original array:\n");
    for (i = 0; i < size; i++)
        printf("%d ", arr[i]);
    printf("\n");

    mergeSort(arr, 0, size - 1);

    printf("Sorted array:\n");
    for (i = 0; i < size; i++)
        printf("%d ", arr[i]);
    printf("\n");

    return 0;
}
```

Output:-

```
srinjoy@srinjoy-HP-250-G8-Notebook-PC: ~/Desktop/DAA_MCA
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ gcc merage_sort.c
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$ ./a.out
Original array:
38 27 43 3 9 82 10
Sorted array:
3 9 10 27 38 43 82
srinjoy@srinjoy-HP-250-G8-Notebook-PC:~/Desktop/DAA_MCA$
```

Q13.Find the cost of MST using given Graph



Ans:-

<u>Code:-</u>

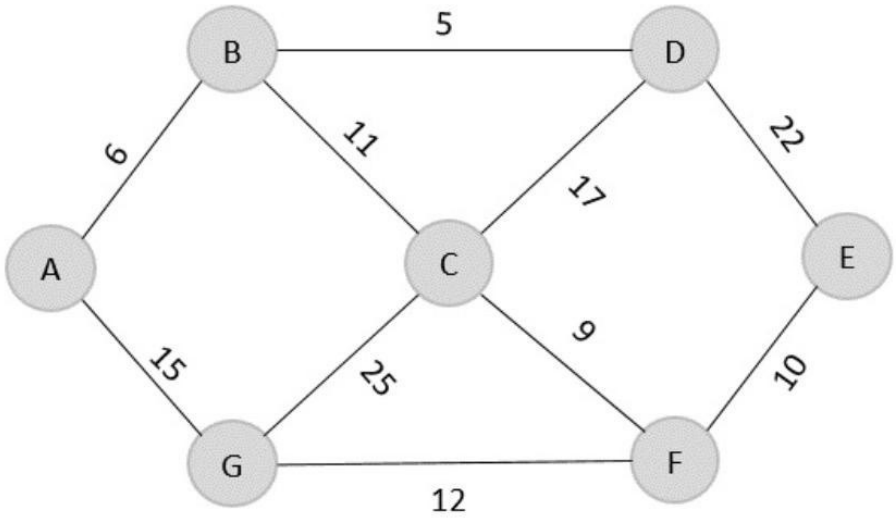
```
#include <stdio.h>

#define u 999

int main() {
    int n = 7;
    int cost[7][7] = {
        {0,1,5,u,u,u,u},
        {1,0,4,8,7,u,u},
        {5,4,0,6,u,2,u},
        {u,8,6,0,11,9,u},
        {u,7,u,11,0,3,10},
        {u,u,2,9,3,0,12},
        {u,u,u,u,10,12,0}
    };
```

```
    int visit[7] = {0};
    int mincost = 0;
    visit[0] = 1;

    for (int e = 0; e < n - 1; e++) {
        int min = u, o = 0, v = 0;

        for (int i = 0; i < n; i++) {
            if (visit[i]) {
                for (int j = 0; j < n; j++) {
                    if (!visit[j] && cost[i][j] < min) {
                        min = cost[i][j];
                        o = i;
                        v = j;
                    }
                }
            }
        }

        visit[v] = 1;
        mincost += min;
        printf("Edge %d-%d cost = %d\n", o, v, min);
    }

    printf("Minimum cost = %d\n", mincost);
    return 0;
}
```

Output

```
devildemon@devildemon-Virtua
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ gcc prim_algo_MST.c
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ ./a.out
Edge 0-1 cost = 1
Edge 1-2 cost = 4
Edge 2-5 cost = 2
Edge 5-4 cost = 3
Edge 2-3 cost = 6
Edge 4-6 cost = 10
Minimum cost = 26
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$
```

Q14. Write a C program that uses Kruskal's algorithm to find and display the minimum spanning tree (MST) of this graph and print the total weight of the MST ?



Ans:-

<u>Code:-</u>

```
#include <stdio.h>
struct edge {
    char u, v;
    int w;
};
int parent[10];

int find(int i) {
    while (parent[i])
        i = parent[i];
    return i;
}
void uni(int i, int j) {
    parent[j] = i;
}
int main() {
    struct edge e[10] = {
```

```
        {'B','D',5}, {'A','B',6}, {'C','F',9}, {'F','E',10},
        {'B','C',11}, {'G','F',12}, {'A','G',15},
        {'C','D',17}, {'D','E',22}, {'C','G',25}
    };

    int i, count = 0, sum = 0;

    printf("Edges in the Minimum Spanning Tree:\n");

    for (i = 0; i < 10 && count < 6; i++) {
        int u = find(e[i].u - 'A');
        int v = find(e[i].v - 'A');

        if (u != v) {
            uni(u, v);
            printf("%c - %c : %d\n", e[i].u, e[i].v, e[i].w);
            sum += e[i].w;
            count++;
        }
    }

    printf("Total weight of MST = %d\n", sum);
    return 0;
}
```

Output

```
                                                    devildemon@devildemon-VirtualBox
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ gcc Kruskal_algo_MST.c
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ ./a.out
Edges in the Minimum Spanning Tree:
B - D : 5
A - B : 6
C - F : 9
F - E : 10
B - C : 11
G - F : 12
Total weight of MST = 53
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$
```

Q15. A set of jobs is given where each job takes one unit of time to complete. Each job has a deadline and an associated profit. The objective is to schedule the jobs in such a way that the total profit is maximized, subject to the constraint that a job must be completed on or before its deadline.

The details of the jobs are as follows:

Jobs	A_1	A_2	A_3	A_4	A_5
Deadlines	2	1	3	2	1
Profit	60	100	20	40	20

Write a C program and display the selected job sequence to with deadlines algorithm using the Greedy approach.Also calculate the maximum profit obtained?

Ans:-

<u>Code:-</u>

```
#include <stdio.h>
struct job {
    char id;
    int deadline;
    int profit;
};
int main() {
```

```
struct job j[5] = {
    {'A',2,60},
    {'B',1,100},
    {'C',3,20},
    {'D',2,40},
    {'E',1,20}
};

int n = 5, max = 0;
for (int i = 0; i < n - 1; i++) {
    for (int k = i + 1; k < n; k++) {
        if (j[i].profit < j[k].profit) {
            struct job temp = j[i];
            j[i] = j[k];
            j[k] = temp;
        }
    }
}
for (int i = 0; i < n; i++)
    if (j[i].deadline > max)
        max = j[i].deadline;

char slot[max];
for (int i = 0; i < max; i++)
    slot[i] = '-';
```

```
    int totalProfit = 0;
    for (int i = 0; i < n; i++) {
        for (int k = j[i].deadline - 1; k >= 0; k--) {
            if (slot[k] == '-') {
                slot[k] = j[i].id;
                totalProfit += j[i].profit;
                break;
            }
        }
    }

    printf("Job sequence: ");
    for (int i = 0; i < max; i++)
        printf("%c ", slot[i]);

    printf("\nTotal Profit = %d\n", totalProfit);

    return 0;
}
```

Output

```
devildemon@devildemon-Virt
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ gcc job_sequence.c
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$ ./a.out
Job sequence: B A C
Total Profit = 180
devildemon@devildemon-VirtualBox:~/Desktop/MINT BAK/DAA_MCA$
```